

Alpilla Enerji Üretim A.Ş.

Alpilla 600 MW Combined Cycle Power Plant

Geotechnical Baseline Summary

Document number	ALP-OWN-GEO-001
Revision	0
Date	2026-09-05
Status	Issued for tender / contract (for information; Contractor to verify)
Confidentiality	Restricted - Contract ALP-EPC-001 parties only

	Prepared	Checked	Approved
Function	Owner's Engineer (discipline lead)	Owner's Engineer (project engineer)	Owner's Project Director
Signature / date	signed 2026-09-05	signed 2026-09-05	signed 2026-09-05

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CASE STUDY - all organisations, data and signatures are fictional.

1. Investigation

12 boreholes (30–45 m), 20 CPTu, 6 MASW profiles, laboratory testing (2025). Grade +15.0 m after Owner's levelling fill (1.5–3.0 m engineered fill, compacted to 95 % Standard Proctor).

2. Stratigraphy (typical)

Depth below grade	Layer	Key parameters
0–2.5 m	engineered fill (sandy gravel)	γ 20 kN/m ³
2.5–9 m	loose to medium dense silty sand (alluvium)	SPT N 6–18; liquefiable under DD-2
9–18 m	soft to firm clay with sand lenses	cu 30–70 kPa, compressible
18–25 m	dense sand and gravel	SPT N > 40
> 18–25 m	weathered to moderately weathered sandstone/marl	UCS 5–25 MPa

- Groundwater at +13.5 m (1.5 m below grade), tidal influence negligible, chloride 3,000–8,000 mg/l (brackish) — aggressive to concrete (XS/XA2).
- Site class per TBDY 2018: ZD (Vs30 = 240 m/s). Site-specific PGA (DD-2): 0.45 g.

3. Owner's note

Deep foundations (bored piles to the sandstone) or ground improvement are expected for all heavy and vibrating structures. The baseline is for information; the Contractor is responsible for interpretation and confirmatory investigation (Sub-Clause 4.5).